

END SEMESTER ASSESSMENT (ESA) - Jan - May 2024**UE20CS204 - Web Technologies****Total Marks : 100.0**

1.a. Create a Placement form with the following controls:

- a. A text box to collect the student's name and SRN
- b. A multiline input field to collect college name and address
- c. A student should give choice to tick their department from a list of departments
 - i. CSE
 - ii. ECE
 - iii. MECH
- d. A collection of three radio buttons for Semester option that are labelled as follows:
 - i. IV
 - ii. V
 - iii. VI
- e. Submit and reset Button (8.0 Marks)

1.b. Describe the structure of HTTP Request and Response message using an example. (6.0 Marks)

1.c. With a neat diagram, explain the CSS Box model and its significance. (6.0 Marks)

2.a. What is Geo Location? Explain the two methods of navigator.geolocation object?

Write a program to display your current location on the browser when the button "Get Current Position" is clicked as shown below.

Get Current Position

Your current location is (Latitude: 12.8615402, Longitude: 77.6642808)

(7.0 Marks)

2.b. Give the difference between JSON and XML.

Represent the details of three 6th sem students both in JSON and XML. Consider the attributes id, name, branch, CGPA (6.0 Marks)

2.c. A HTML page contains the following:

- A table exists with id "table1"
- A button "Add" with id "btn1"
- A div with id "display"

Add JavaScript to the webpage for the following specifications

- On clicking "Add" a new row with 2 cells are added to the table
- Populate each cell with a random number (between 1 and 200)
- If the cell contains an even number, mouse over should turn the cell green
- If the cell contains an odd number, mouse over should turn the cell red.
- If you click on a cell, the div will be populated with the cell content

Note: Write only vanilla JavaScript code. Assume that body tag has an onload="init()" handler.

(7.0 Marks)

3.a. With a neat diagram explain component life cycle.

(8.0 Marks)

3.b. Create a class component which will render a H1 and paragraph element apply different inline style for H1 and P element created using java script objects (key value pair). (7.0 Marks)

3.c. What is the significance of key property? Describe with an example. (5.0 Marks)

4.a. Write in brief the important features of Node JS. (6.0 Marks)

4.b. Write JavaScript code using NodeJS modules only (not Express JS) to accept a GET request in the form of `http://server:8000/calc?op=add&op1=10&op2=20` and returns back the calculated output, in this case the addition of 10 and 20, which is 30. The operations supported by this basic calculator are addition, subtraction, multiplication and division. (8.0 Marks)

4.c. What are buffers in Node.js? With help of syntax explain any 2 buffer operations. (6.0 Marks)

5.a. What is RESTful API? Explain any 4 design specification/constraints of REST API. (6.0 Marks)

5.b. What are Middleware Functions and what are they used for? Explain Router-level middleware with an example. (8.0 Marks)

5.c. When doing routing of URLs in ExpressJS, we used routes that contained a colon character (e.g. `"/hello/:id"`) yet we never included a colon in the hierarchical part of a URL we used.

(i) Explain the purpose of this colon character?

(ii) Describe what would happen if we just deleted the colon from the routes.

(6.0 Marks)